

Generation Interconnection Request System Impact Study Report

For Queue Project Q18-1067

Prairie Horizon Agri Energy 138 kV
May 23, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1067. The Interconnection Customer proposes a 305.6 MW solar photovoltaic facility interconnecting at the Prairie Horizon Agri Energy 138 kV substation in the Kentucky area. The request was received on September 23, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 336 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$23,569,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Usaa 1 - Prairie Horizon Agri Energy 138 kV	Prairie Horizon Agri Energy - Ogdensburg 138 kV	112.4	227	53.6
Damon - E Broad 138 kV line	Spatsum Lornex - Latah Jct 138 kV	112.2	217	19.2
Usaa 1 - Greenville Ave 138 kV Line	Prairie Horizon Agri Energy - Ogdensburg Substation 138 kV	109.6	223	53.6
Russellville E - Rancho Arnel 138 kV Line	Albion Road - Platte City 138 kV L	104.7	342	7.6
Holden - Embudo 138 kV L	Lee County Solid Waste Energy - Cooper 345 kV	104.3	236	10.9
Santa Rosa Beach - Tule Substation 138 kV L	N Appleton - Davisville 138 kV	103.4	239	6.3
Santa Rosa Beach - West Mount Vernon 138 kV	N Appleton - Davisville 138 kV	103.0	254	6.3
Nsp - W Mount Vernon 138 kV	Davisville - North Appleton 138 kV Line	102.6	270	6.3
Wissota - Nsp 138 kV	Davisville Substation - N Appleton 138 kV L	102.2	285	6.3
Minco - Barrett 230 kV	Barrett - Norwich 500 kV line	101.4	620	7.8
Berr St 345-138 kV TX	Omni-Hackberry - Nrwich Substation 500 kV	100.9	547	11.6
Ironman - Goldsboro 230 kV L	Norwich - Omni-Hackberry 500 kV	100.1	536	6.5
Union - Garden Villas 138 kV	Union - Desoto Pleasant Run 138 kV	100.0	194	5.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Prairie Horizon Agri Energy - Usa 1 138 kV L	Rebuild line with higher-capacity conductor	\$12,682,000	69% (\$8,751,000)
East Broad - Damon 138 kV Line	Rebuild line with higher-capacity conductor	\$2,363,000	42% (\$987,000)
Usaa 1 - Greenville Ave 138 kV Line	Reconductor line	\$2,401,000	69% (\$1,657,000)
Russellville E - Rancho Armel 138 kV Line	Reconductor line	\$5,264,000	12% (\$621,000)
Embudo - Holden 138 kV	Reconductor line	\$3,216,000	17% (\$546,000)
Santa Rosa Beach - Tule 138 kV	Reconductor line	\$2,188,000	12% (\$255,000)
West Mount Vernon - Santa Rosa Beach Substation 138 kV	Reconductor line	\$1,625,000	12% (\$190,000)
Nsp - W Mount Vernon 138 kV	Reconductor line	\$3,175,000	12% (\$371,000)
Wissota - Nsp 138 kV	Reconductor line	\$8,442,000	12% (\$986,000)
Minco - Barrett 230 kV	Reconductor line	\$4,599,000	14% (\$638,000)
Berr St 345-138 kV TX	Replace transformer with higher-rated unit	\$22,939,000	18% (\$4,131,000)
Goldsboro - Ironman 230 kV	Reconductor line	\$26,022,000	13% (\$3,451,000)
Garden Villas Substation - Union 138 kV	Reconductor line	\$7,392,000	13% (\$985,000)
	Total allocated to project		\$23,569,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.